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LECTURE NOTES

# Kernel Methods

AI512 — Introduction to Machine Learning

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2026-09-04



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## 1 Transforming the input space

Suppose that we have a dataset where we can't easily classify data using a support vector (4. Support Vector Machines (SVM)). We can transform the space using a transformation function, to model the data by using more dimensions. Take the example from the lectures

$$T(x, y) = (x^2, y^2, \sqrt{2}xy)$$

This way we can use a hyperplane of +1 dimensionality to classify the data.

## 2 Kernel Trick

Suppose that  $T(x)$  is a mapping to a higher dimension

Note that algorithms like SVM only need **dot products**

That means that with

$$k(x_i, x_j) = \phi(x_i)^T \phi(x_j)$$

Because of the definition of the inner product, we can describe the relation between two points using  $k(x_i, x_j)$

Computing  $\phi(x)$  explicitly can be **very expensive or infinite-dimensional**

- **Linear Kernel:**  $k(x, y) = x^T y$
- **Polynomial Kernel:**  $k(x, y) = (x^T y + c)^d$
- **Radial Basis Function (RBF) Kernel:**  $k(x, y) = \exp(-\frac{\|x-y\|^2}{2\sigma^2})$
- **Sigmoid Kernel:**  $k(x, y) = \tanh(\gamma x^T y + c)$

## 3 Kernel regression

(no notes yet)

## 4 Support Vector Machines (SVM)

Among many possible hyperplanes that may separate the data points of two classes, SVM chooses the one that maximizes the smallest distance between a data point and the hyperplane.

This distance is called the **margin**. The data points that are closest to the hyperplane are called **support vectors**.

Note that we use because of these property, SVM is also called a **maximum margin classifier**. We would normally use k-fold cross validation, and balance the bias-variance tradeoff to ensure a maximum margin.

In this example is 2d, where the support vectors are lines